

BENGALURU TRAFFIC INTELLIGENCE PLATFORM

Executive Report — Final Submission

Generated: 19 June 2026, 17:59 IST | Gridlock Hackathon 2026

COMMAND CENTER METRICS

Metric	Value	Source
Total Incidents Analyzed	10	demo_results.csv
Critical Incidents (75-100)	4 (40%)	demo_results.csv
Officers Required	127	demo_results.csv
Barricades Required	183	demo_results.csv
Patrol Vehicles	38	demo_results.csv
Average Delay	3.0 min	demo_results.csv
Avg Congestion Reduction	20.5%	Computed: risk_tier × score
Avg Delay Reduction	69.3%	Computed: delay vs alt route
Total Vehicles Diverted	7,992	Computed: corridor vol × score
Highest Risk Corridor	ORR East 1 (95.0/100)	demo_results.csv

SCENARIO-BY-SCENARIO BREAKDOWN

Scenario	Risk	Score	■	■	Corridor	Delay↓	Cong↓	Diverted
S01 Accident ORR	CRITICAL	95	13	18	ORR East 1	56%	36%	1,463
S02 Vehicle Breakdown Hosur Rd	MODERATE	53	4	8	Hosur Road	43%	11%	553
S03 Waterlogging Whitefield	CRITICAL	85	9	13	ORR East 2	77%	32%	1,168
S04 Protest Town Hall	CRITICAL	85	26	33	CBD 1	80%	32%	1,028
S05 VIP Movement Bellary Rd	CRITICAL	82	22	26	Bellary Road 1	69%	31%	947
S06 Metro Construction ORR	MODERATE	53	6	19	ORR East 2	90%	11%	728
S07 IPL Match Chinnaswamy	MODERATE	48	15	19	CBD 2	50%	10%	528
S08 Tree Fall Sankey Road	HIGH	66	6	12	Bellary Road 1	83%	18%	762
S09 Procession Mysore Road	HIGH	59	14	20	Mysore Road	56%	16%	584
S10 Public Gathering Lalbagh	MODERATE	35	12	15	Non-corridor	90%	7%	231

CORRIDOR RISK RANKING

Rank	Corridor	Avg Score	Risk Tier
1	ORR East 1	95/100	CRITICAL
2	CBD 1	85/100	CRITICAL
3	Bellary Road 1	74/100	HIGH
4	ORR East 2	69/100	HIGH
5	Mysore Road	59/100	HIGH
6	Hosur Road	53/100	MODERATE
7	CBD 2	48/100	MODERATE
8	Non-corridor	35/100	MODERATE

AI JUDGE SUMMARY (Gemini 2.5 Flash)

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Judge Summary — 19 June 2026, 17:59 IST

1. PLATFORM OVERVIEW

The Bengaluru Traffic Intelligence Platform autonomously detects, scores, and responds to 10 real-world traffic events across key city corridors. It deploys alternate routes, quantifies resource needs, and generates AI-driven incident reports using Gemini 2.5 Flash.

2. KEY METRICS (from demo_results.csv)

- Incidents analyzed : 10
- Critical incidents : 4 (40%)
- Officers recommended : 127
- Barricades required : 183
- Patrol vehicles : 38
- Average delay : 3.0 min
- Avg congestion reduction: 20.5%
- Avg delay reduction : 69.3%
- Total vehicles diverted : 7,992

3. TOP-RISK CORRIDORS

1. ORR East 1 — Score 95/100
2. CBD 1 — Score 85/100
3. Bellary Road 1 — Score 74/100

SYSTEM ARCHITECTURE

Component	Technology
Road Network	OSMnx 2.1.0 + NetworkX
Visualization	Folium 0.20.0 (CartoDB dark tiles)
AI Reports	Gemini 2.5 Flash
Congestion Scoring	Rule-based engine (0-100 scale)
Resource Recommender	Cause × Priority × Corridor × Rush Hour
Dataset	8,173 Bengaluru traffic events (Astram anonymized)
ML Status	No ML training — target leakage audit passed

